

Algebra 1
Unit 2
Lesson 2 Homework

Name _____
Date _____
Period _____

1. The **quotient of powers law** of exponents states keep the then _____ the powers.

add
subtract
multiply

base
coefficient
exponent

2. The **power of a quotient law** of exponents states the exponential expression $(\frac{a}{b})^m$ is equal to

- ☐ a^{b-m} .
☐ $\frac{a^m}{b^m}$.
☐ $a^m - b^m$.

Find the quotient of each expression by using the laws of exponents to generate an equivalent expression.

3. $\frac{-4x^4y^2}{10xy^7}$

4. $\frac{3a^{10}b^6}{6b^2c}$

5. $(\frac{-2r^3s}{3t^4})^6$

6. $\frac{-k^7j^3}{7k^9j^3}$

$$7. \quad \frac{46w^3z^{10}}{33x^2y^{11}}$$

$$8. \quad \frac{-144c^9d^2e^0}{98c^4d^{11}e}$$

$$9. \quad \left(\frac{z^0}{-3x^3}\right)^4$$

$$10. \quad \frac{5s^{15}t^6}{-10s^{12}t}$$